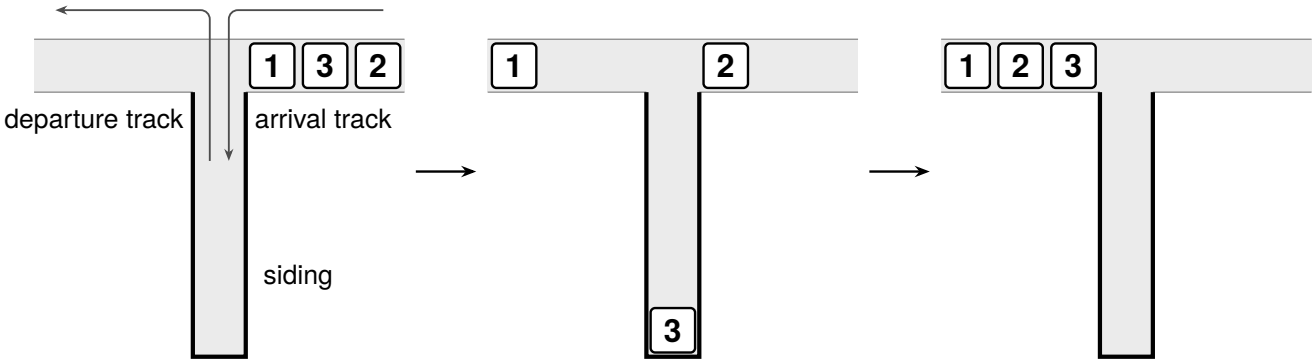
Line up a train on the arrival track the way it is drawn. The car on the left goes first, and cars never jump over each other. Each car drives down into the siding and then out to the departure track. Only the top car in the siding can drive out. Car 1 must leave first, then car 2, then car 3, and so on.



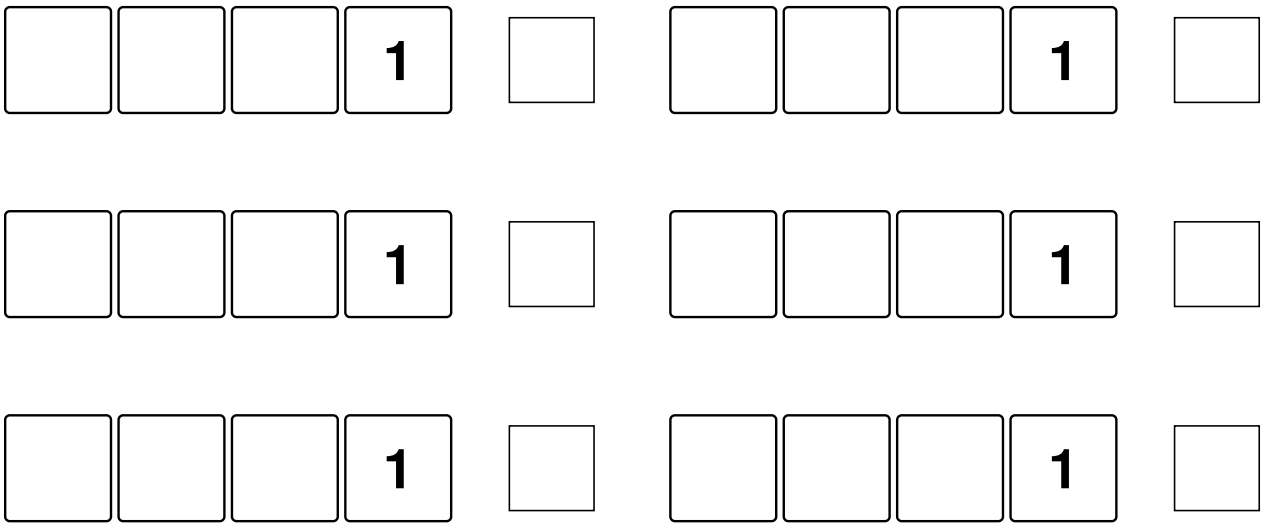
Problem 1: Which of these trains can leave 1, 2, 3? Mark each one ✓ or ✗, and show with the cars why each ✗ train gets stuck.

1 2 3	☐	1 3 2	☐
2 1 3	☐	2 3 1	☐
3 1 2	☐	3 2 1	☐

Problem 2: Which of these trains can leave 1, 2, 3, 4? Mark each one ✓ or ✗.

2	1	4	3	☐	3	4	1	2	☐
4	3	2	1	☐	1	4	3	2	☐
4	2	3	1	☐	3	1	2	4	☐
4	1	3	2	☐	1	3	4	2	☐

Problem 3: Car 1 is last in each of these trains. Fill in cars 2, 3, 4 in different ways, and find every train that can leave 1, 2, 3, 4.



Problem 4: Which of these trains can leave 1, 2, 3, 4, 5? Mark each one ✓ or ✗.

4	5	1	2	3	☐	2	1	5	4	3	☐
5	4	3	2	1	☐	5	1	3	2	4	☐
1	3	5	2	4	☐	3	2	5	1	4	☐

Problem 5: One partner lines up all five cars, and the other partner guesses whether they can leave 1, 2, 3, 4, 5. Drive them to check, then switch jobs.

Problem 6: Find all 10 trains of cars 1, 2, 3, 4 that cannot leave 1, 2, 3, 4.

